Graph-Based Recommenda

CP4285 Modern Recommendation Systems — Week 09

CP4285 Instruction Team

13 Oct 2026

Scan for the slides
soc-n.us/cp4285-t2610-w09

Bandit Time: Start TempoQuiz

Week 08 recap

Join using the live host-dashboard QR or room code.

Five minutes including joining and explanations.

What does a preference pair tell us?

Recall BPR, ranked-list utility and exposure.

The instructor will release the questions in TempoQuiz.

WEEK 08 RECAP

One ranked list, three learning targets

Pointwise

Is this item relevant?

Learn a score or label for each user–item pair independently.

Useful when explicit and calibrated scores are given.

Pairwise

Which of two items comes first?

Learn that an observed or preferred item should outrank a comparison item.

Useful when relative preference is clearer than a rating.

Listwise

How should the candidate group rank?

Define a loss using the candidate group, such as a target distribution.

Position-sensitive metric alignment depends on the training loss and update weighting rules.

BPR: pairwise ranking from implicit feedback

$$T = \{(u, i, j) : i \in I_u,\ j \notin I_u\}, \qquad \Delta_{uij} = s(u, i) - s(u, j)$$

Here I_u contains the items user u interacted with. **T defines assumed training preferences**, not proven dislikes.

Objective versus model

BPR specifies the training criterion. BPR-MF uses matrix factorisation to supply the score.

Other differentiable scoring models can use the same pairwise loss.

Observed does not mean universally preferred

A sampled unobserved item may be unseen, not disliked. The sampler determines which comparisons training encounters.

Listwise quality: measure the displayed order

Relative order, not position weighting: basic BPR has no explicit NDCG swap weights.

1 · Convert grades to gains

$g_i = 2^{y_i} - 1$

Company	Grade	Gain computation
Akari	1	$2^1 - 1 = 1$
Banyan	2	$2^2 - 1 = 3$
Chandra	0	$2^0 - 1 = 0$
Dempsey	3	$2^3 - 1 = 7$

Gains come from assessed grades. Dempsey contributes zero to the current DCG@3 because it is at rank 4.

2 · Discount gains by position

$DCG@3 = \sum_{r=1}^{3} \frac{g_r}{\log_2(r+1)}$

Rank 1/2/3 discounts: 1, 0.631, 0.5.

Current: Akari → Banyan → Chandra

$DCG = 1(1) + 3(0.631) + 0(0.5)$
$= 2.893$

Ideal: Dempsey → Banyan → Akari

$IDCG = 7(1) + 3(0.631) + 1(0.5)$
$= 9.393$

3 · Normalize

$NDCG@3 = \frac{2.893}{9.393} \approx 0.308$ · **Ideal = 1.**

BPR in the learning-to-rank map

Method	Training signal	Correction
Pointwise	One item and its label	Item-level prediction error
RankNet	Relevance preference pair	Logistic pairwise error
BPR	Observed / unobserved user pair	Logistic error + pair sampling
LambdaRank	Pair in the current ranking	Pairwise signal × swap impact
ListNet	Candidate-group distribution	Group cross-entropy

BPR-MF = a matrix-factorisation scorer trained with BPR. Neither a neural scorer nor BPR alone specifies fair exposure.

01 Announcements and Context

Changelog

- **13 Oct 2026** — Added graph walks, priors, learned propagation, worked music examples and two activity cycles. Replaced S10 with six verified Canvas quotations and author avatars grouped by three evidence considerations.
- **6 Oct 2026** — Added TempoQuiz, exercise review and recommendation-history break.
- **5 Oct 2026** — Added the next-week prelecture exercise and preview.



ANNOUNCEMENTS

Course checkpoints

Essay 2: deadline was Mon, 12 Oct · 23:59 SGT.

TEAMMATES: indicative feedback has been released.
This feedback is not counted towards your grade.

STePS: all teams except **3 and 6** submitted that they will present.

Weeks 10 and 11: optional **15-minute sessions** for additional project feedback.

Presentation deck critiques: to be released later this week.



PROJECT ADVICE

Make experiments explain

Justify your choices: why this model, dataset and evaluation metric?

Explain successes and failures: understanding matters more than complexity or state-of-the-art results.

Look beyond an aggregate score: inspect individual examples and meaningful subgroups.

Build on prior work: read, cite and, where feasible, replicate it. Contribute your own analysis.



EXPERIMENTATION ADVICE

Ablations and experiment scale

Test a component's contribution

Predict the effect before running the experiment.

Remove a feature group or component; additive comparisons also help.

Retrain each variant. Keep the split and evaluation protocol fixed.

Explain where results change and why.

Validate before scaling

1–3 min · Toy: check correctness on tiny data.

30–60 min · Medium: test promising ideas and inspect results.

Overnight · Full: run validated settings at scale.

Estimate runtime from initial runs; sample data to fit your budget.



Learning outcomes

- Represent recommendations as graph problems.
- Explain graph propagation.
- Build graph-based recommenders.
- Analyze risks of graph-driven feedback loops.



LAST WEEK PRE-LECTURE EXERCISE · W08

Popularity or connection? Fix the surface

Canvas · Solo · 15 minutes · no peer reply

Complete by Mon, 12 Oct 2026 · 23:59 SGT

Choose **one surface** and keep its user, task, and constraints fixed throughout your answer.

A · Music autoplay

Next track after an acoustic-folk playlist. Keep the genre transition gentle.

B · Jobs search

First search-result card for a graduate seeking entry-level data work in Singapore.

C · Grocery checkout

One “Complete your meal” checkout item for a vegetarian dinner basket. It must be available.

Both candidates are eligible: X has 1,000 interactions but no known overlap with this user’s interests; Y has 20 interactions, several from users with overlapping histories. Counts come from the same period and interaction definition.

Post 120–160 words: name your surface and concrete X/Y items; choose which to investigate first; state one assumption, one check of exposure, and evidence that would reverse your choice.

Today’s lecture: ask what the pattern of connections reveals beyond an item’s total count.

LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

What evidence lies behind the count?

Relevant connections

Does overlap reflect the same needs?

“The overlapping history would not be useful as a connection if the overlapping histories were about something else like toiletries.”

CONTEXT OF THE OVERLAP



“My assumption is that similar baskets reflect similar meal needs, and that both products meet the shopper’s dietary requirements.”

PREFERENCES AND ELIGIBILITY



Exposure & visibility

Did placement produce the count?

“I would check exposure by looking at how often each job was shown to similar users in comparable search-result positions, then compare their interaction rates.”

COMPARABLE OPPORTUNITIES



“X may have accumulated more interactions simply because it received greater exposure.”

VISIBILITY BEHIND POPULARITY



Interaction meaning

What does an observed edge mean?

“Assumption: Interactions are positive signals (full plays, saves), not skips or accidental plays.”

POSITIVE INTEREST OR INCIDENTAL ACTIVITY?



“What would reverse my choice is if Y’s 20 interactions turn out to be mostly apply-then-reject or quick bounces, or if X’s clickers include lots of similar grads after all.”

EVIDENCE THAT REVERSES A CHOICE



Before propagating an edge, what evidence does that edge carry?

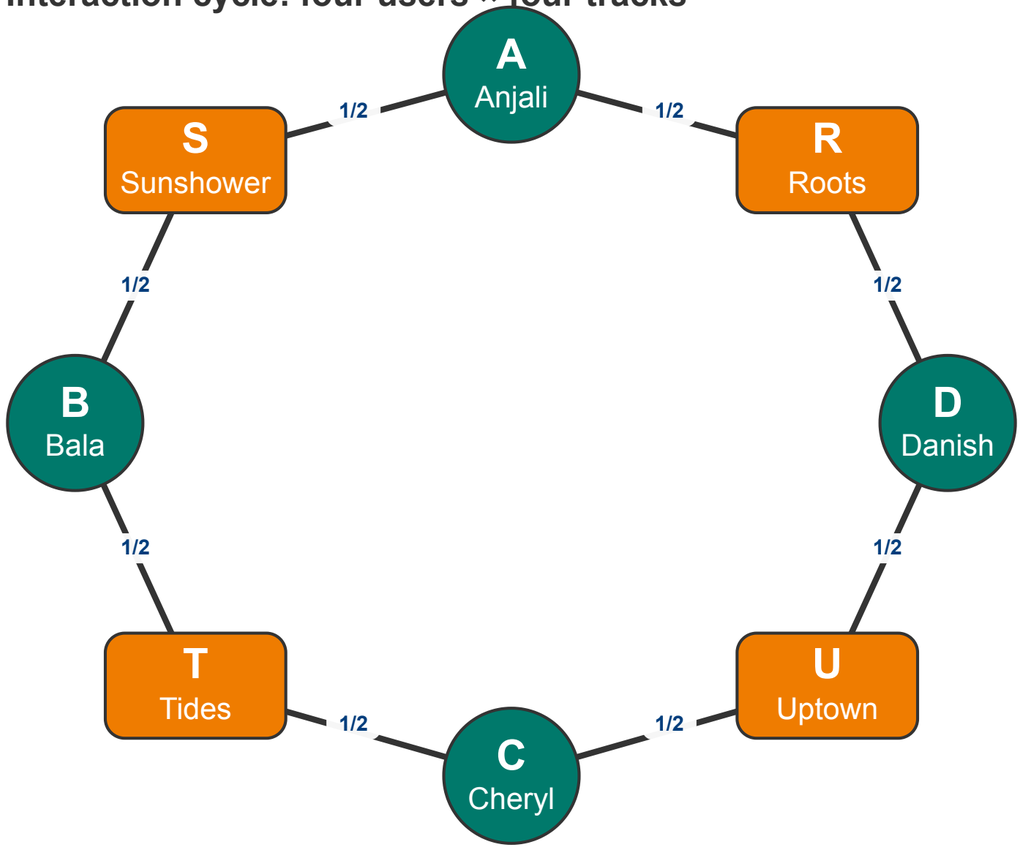
02 Ranking Through Graphs

A shared example for walks, priors and learned propagation.



Four listeners, four tracks

Interaction cycle: four users × four tracks



Users = teal circles; tracks = orange rounded rectangles. Every node has degree 2.

Listener	Observed tracks
A · Anjali	R, S
B · Bala	S, T
C · Cheryl	T, U
D · Danish	U, R

R **Roots** · world fusion
S **Sunshower** · dance pop
T **Tides** · instrumental surf rock
U **Uptown** · funk pop

Anjali's unseen candidates: **T, U**.

Degrees and paths

Music interactions are undirected: either direction is allowed.

Every node has degree **2**.

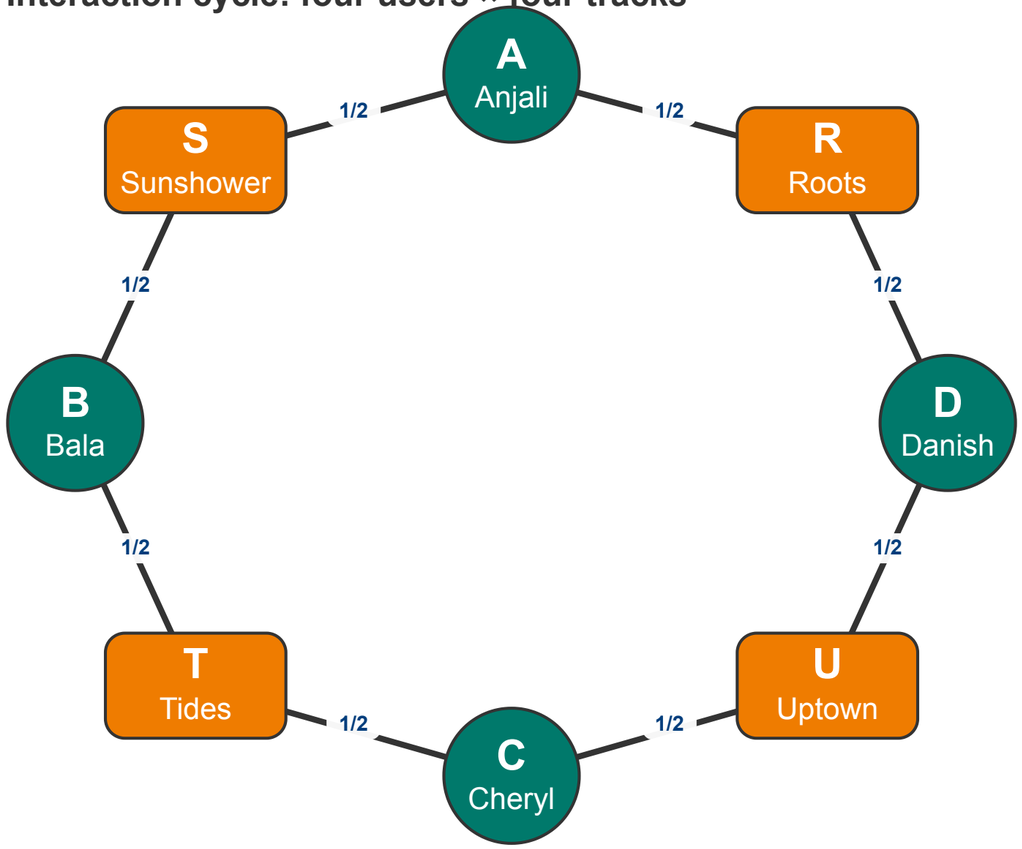
Each direction uses its **source degree**:

$$P_{ij} = \frac{1}{d_i} = \frac{1}{2}.$$

Anjali → Sunshower → Bala → Tides
Anjali → Roots → Danish → Uptown

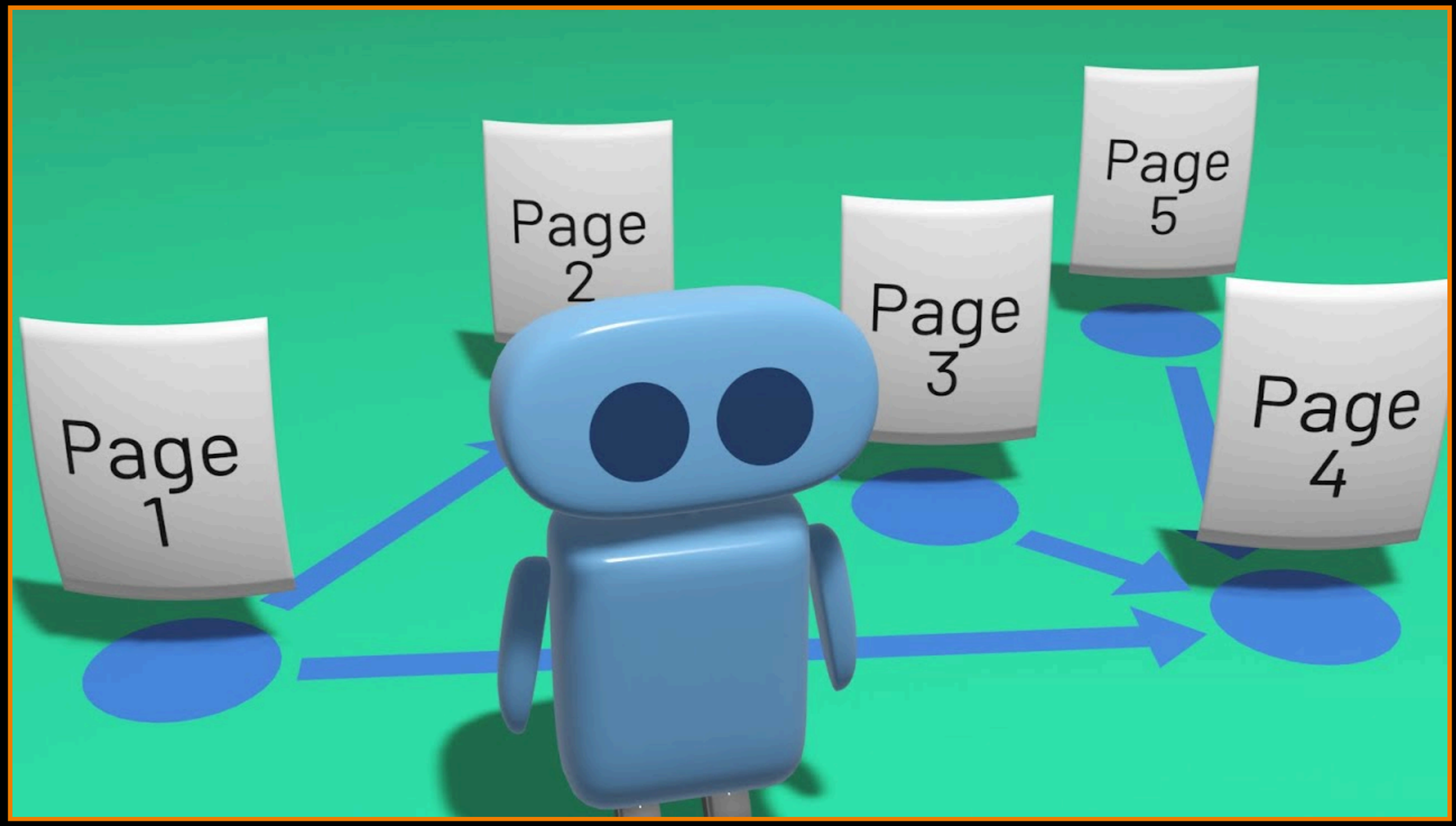
If Anjali had degree 2 and Roots degree 3:
Anjali → Roots: $1/2$; Roots → Anjali: $1/3$.

Interaction cycle: four users × four tracks



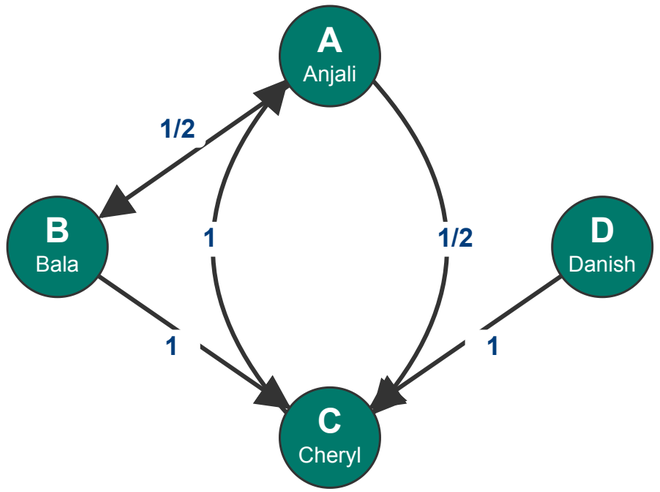
Users = teal circles; tracks = orange rounded rectangles. Every node has degree 2.

Video: PageRank's random surfer



Markov state and PageRank

Same people, new relation: arrows mean follows.

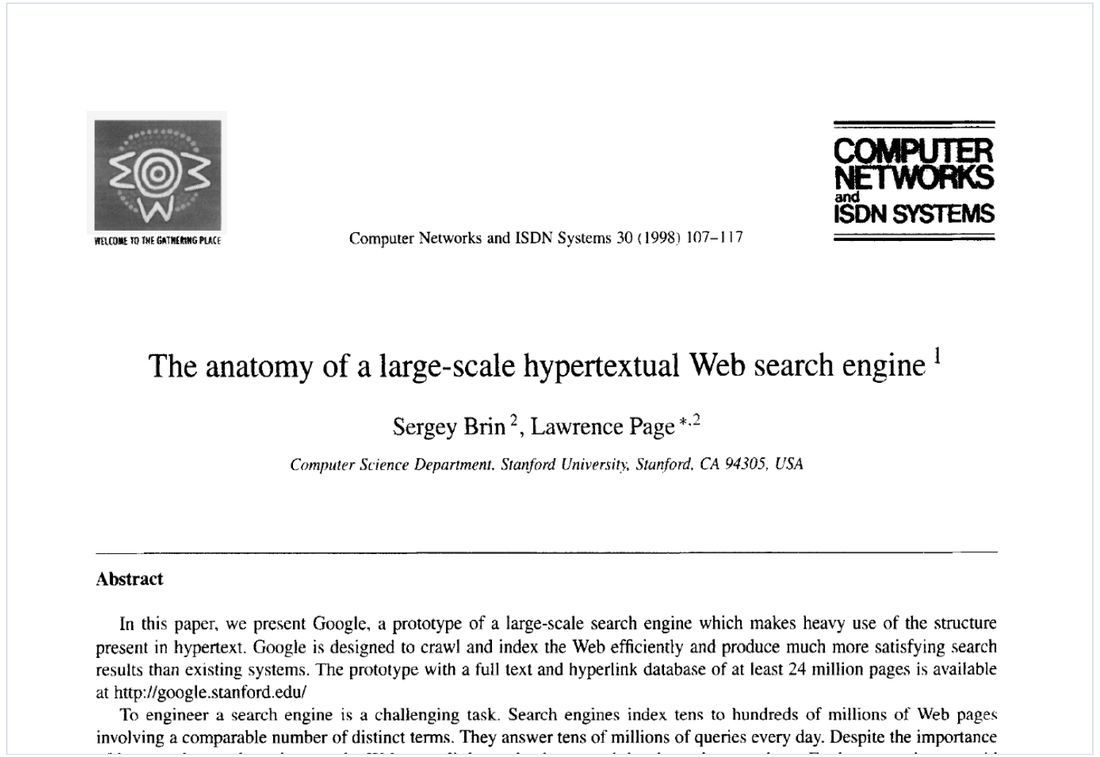


Start at Anjali (A): $r^{(0)} = (1,0,0,0)^\top$; order **A, B, C, D**.

$$r^{(k+1)} = P^\top r^{(k)}.$$

$r^{(1)} = (0, \frac{1}{2}, \frac{1}{2}, 0).$

$r^{(2)} = (\frac{1}{2}, 0, \frac{1}{2}, 0).$



WELCOME TO THE GATHERING PLACE

Computer Networks and ISDN Systems 30 (1998) 107–117

COMPUTER NETWORKS and ISDN SYSTEMS

The anatomy of a large-scale hypertextual Web search engine [1]

Sergey Brin [2], Lawrence Page *,[2]

Computer Science Department, Stanford University, Stanford, CA 94305, USA

Abstract

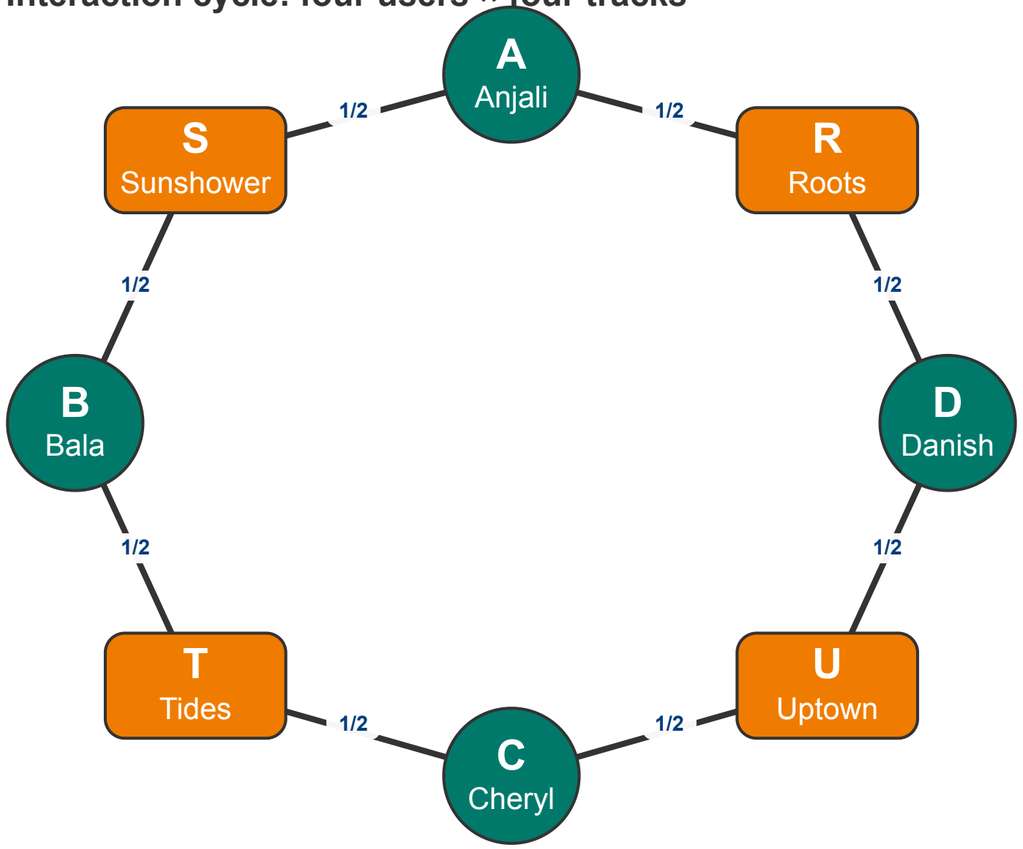
In this paper, we present Google, a prototype of a large-scale search engine which makes heavy use of the structure present in hypertext. Google is designed to crawl and index the Web efficiently and produce much more satisfying search results than existing systems. The prototype with a full text and hyperlink database of at least 24 million pages is available at http://google.stanford.edu/

To engineer a search engine is a challenging task. Search engines index tens to hundreds of millions of Web pages involving a comparable number of distinct terms. They answer tens of millions of queries every day. Despite the importance

Brin, S., & Page, L. (1998). *The anatomy of a large-scale hypertextual Web search engine*. *Computer Networks and ISDN Systems, 30*(1-7), 107–117. https://doi.org/10.1016/S0169-7552(98)00110-X

Stationarity and periodicity

Interaction cycle: four users × four tracks



Users = teal circles; tracks = orange rounded rectangles. Every node has degree 2.

Return to the undirected music cycle.
Starting at Anjali, total user-side mass is:

$$1, \quad 0, \quad 1, \quad 0, \ldots$$

A return to a user takes an even number of steps. The chain has **period 2**.

Yet a stationary distribution exists:

$$\pi_j = \frac{d_j}{\sum_i d_i} = \frac{2}{16} = \frac{1}{8}.$$

Stationarity does **not** imply convergence from every starting state.

Personalised PageRank: compare one step

Back to social follows, order **A, B, C, D**.

One-hot start: $r^{(0)} = \mathbf{1}_A = (1,0,0,0)^\top$.
All initial mass is at **Anjali**.
Restart: probability $\tau = 1/2$; destination distribution q.

$$r^{(1)} = (1-\tau)P^\top r^{(0)} + \tau q.$$

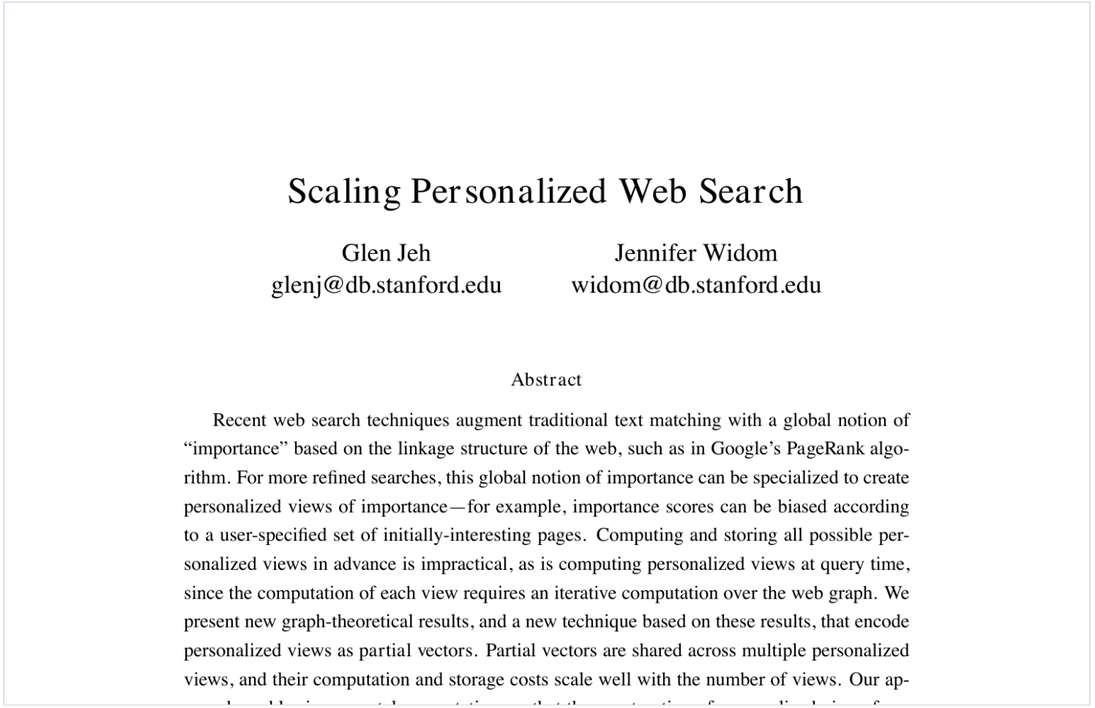
Global PR: $q = (1/4, 1/4, 1/4, 1/4)^\top$.
PPR: $q = \mathbf{1}_A$; restart only at Anjali.

Method	A	B	C	D
Walk only	0	1/2	1/2	0
Global PR	1/8	3/8	3/8	1/8
PPR at A	1/2	1/4	1/4	0

Global at B: $\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{4} = \frac{3}{8}$.

PPR at A: $\frac{1}{2} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2}$.

Each row sums to 1. These are one-step masses.



Scaling Personalized Web Search

Glen Jeh
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Abstract

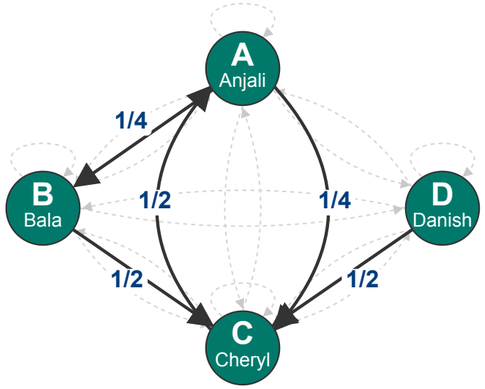
Recent web search techniques augment traditional text matching with a global notion of "importance" based on the linkage structure of the web, such as in Google's PageRank algorithm. For more refined searches, this global notion of importance can be specialized to create personalized views of importance—for example, importance scores can be biased according to a user-specified set of initially-interesting pages. Computing and storing all possible personalized views in advance is impractical, as is computing personalized views at query time, since the computation of each view requires an iterative computation over the web graph. We present new graph-theoretical results, and a new technique based on these results, that encode personalized views as partial vectors. Partial vectors are shared across multiple personalized views, and their computation and storage costs scale well with the number of views. Our ap-

Jeh, G., & Widom, J. (2003). *Scaling personalized web search*. In *Proceedings of the 12th International Conference on World Wide Web* (pp. 271–279). ACM.

Teleportation and long-term probabilities

$$r^* = \frac{1}{2} P^\top r^* + \frac{1}{2} q, \qquad r^* = (a, b, c, d)^\top.$$

Global PR · $q = (1/4, 1/4, 1/4, 1/4)$



1 · Balance · each restart contribution $g = 1/8$:

$$d = g, \qquad b = a/4 + g, \qquad a = c/2 + g$$

$$c = a/4 + b/2 + d/2 + g.$$

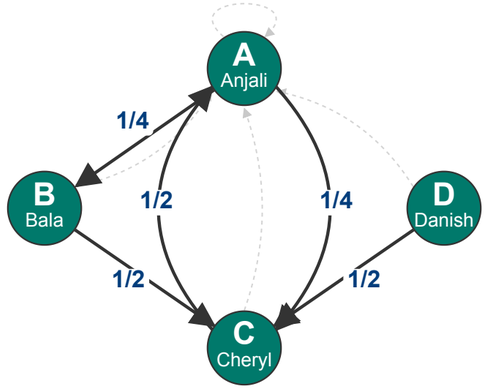
2 · Substitute: $c = 3a/8 + 2g$; $a = 3a/16 + 2g$.
Hence $13a/16 = 1/4$, so $a = 4/13$.

3 · Long-term mass · order A, B, C, D:

$$r^*_{\mathrm{PR}} = \frac{1}{104}(32, 21, 38, 13)^\top.$$

Cheryl > Anjali > Bala > Danish · sum = 1.

PPR at Anjali · $q = (1, 0, 0, 0)$



1 · Balance · restart contributes $1/2$ only to A:

$$d = 0, \qquad b = a/4, \qquad a = c/2 + 1/2$$

$$c = a/4 + b/2.$$

2 · Substitute: $c = 3a/8$; $a = 3a/16 + 1/2$.
Hence $13a/16 = 1/2$, so $a = 8/13$.

3 · Long-term mass · order A, B, C, D:

$$r^*_{\mathrm{PPR}} = \frac{1}{13}(8, 2, 3, 0)^\top.$$

Anjali > Cheryl > Bala > Danish · sum = 1.

👥 Which seed changes the recommendation?

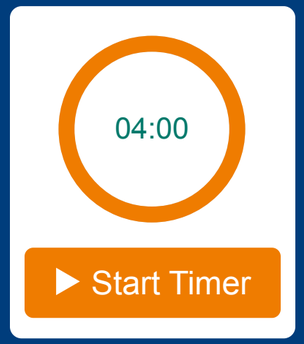
Back to music: item → user → item

Groups of 2–3: start once at **Roots** and once at **Sunshower**. Split mass equally at each node.

Edges: Anjali–R,S; Bala–S,T; Cheryl–T,U; Danish–U,R.

Output: two distributions over R,S,T,U and Anjali's best unseen track.

Average both distributions. What happens to T versus U?



Seed choice and a structural tie

Seed	Roots R	Sunshower S	Tides T	Uptown U
R	1/2	1/4	0	1/4
S	1/4	1/2	1/4	0

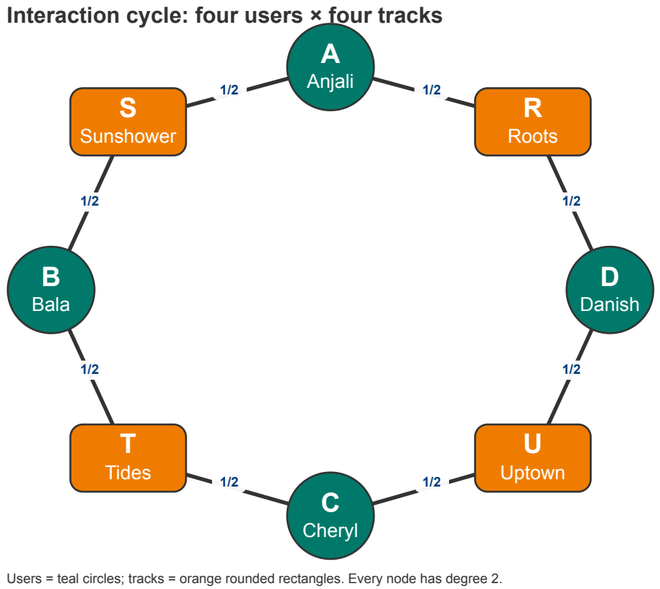
Equal seed: $(3/8,\ 3/8,\ 1/8,\ 1/8)$.

Filter observed **R,S**: **T and U tie because this graph and seed are symmetric**. A finite-depth zero does not establish irrelevance.



One graph, three ranking ideas

Anjali's music graph: same listening links, different ranking mechanisms.



Method	Ranking idea	Output
PageRank / PPR	Walk along links. Global or personal **q** selects where probability mass restarts.	Probability mass Sum = 1
BiRank	Couple **user and item scores**, with separate priors and symmetric normalization.	Two ranking-score vectors
TriRank	Add **explicit aspects** and a third coupled score vector, using review-derived evidence or supplied metadata.	Ranking scores and aspect routes

An aspect route offers an explanation, not proof of relevance.

BiRank: two coupled score vectors

Users u and items v have separate priors.

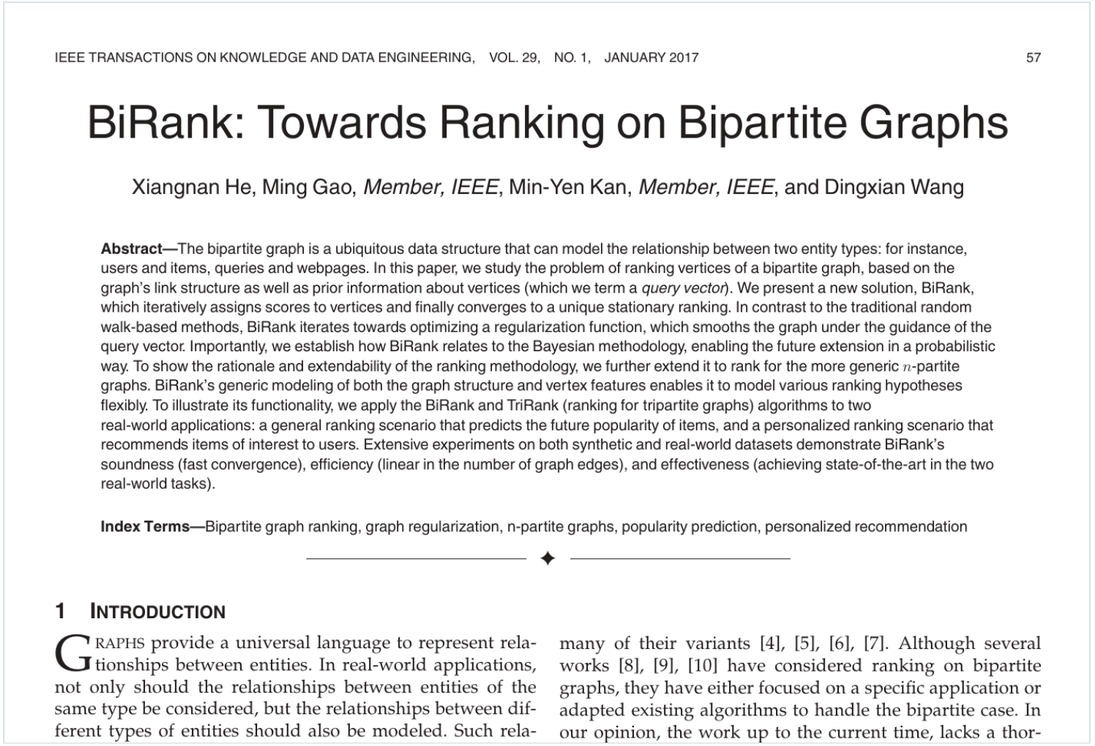
$$S = D_u^{-1/2} W D_i^{-1/2}.$$

$$v^{(k+1)} = \alpha S^\top u^{(k)} + (1-\alpha) v^0.$$

$$u^{(k+1)} = \beta S v^{(k+1)} + (1-\beta) u^0.$$

$u^{(0)} = u^0 = \mathbf{1}_A$, $v^0 = 0$, $\alpha = 1/2$:

$$v_R^{(1)} = v_S^{(1)} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$



IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, VOL. 29, NO. 1, JANUARY 2017 57

BiRank: Towards Ranking on Bipartite Graphs

Xiangnan He, Ming Gao, *Member, IEEE*, Min-Yen Kan, *Member, IEEE*, and Dingxian Wang

Abstract—The bipartite graph is a ubiquitous data structure that can model the relationship between two entity types: for instance, users and items, queries and webpages. In this paper, we study the problem of ranking vertices of a bipartite graph, based on the graph's link structure as well as prior information about vertices (which we term a *query vector*). We present a new solution, BiRank, which iteratively assigns scores to vertices and finally converges to a unique stationary ranking. In contrast to the traditional random walk-based methods, BiRank iterates towards optimizing a regularization function, which smooths the graph under the guidance of the query vector. Importantly, we establish how BiRank relates to the Bayesian methodology, enabling the future extension in a probabilistic way. To show the rationale and extendability of the ranking methodology, we further extend it to rank for the more generic n-partite graphs. BiRank's generic modeling of both the graph structure and vertex features enables it to model various ranking hypotheses flexibly. To illustrate its functionality, we apply the BiRank and TriRank (ranking for tripartite graphs) algorithms to two real-world applications: a general ranking scenario that predicts the future popularity of items, and a personalized ranking scenario that recommends items of interest to users. Extensive experiments on both synthetic and real-world datasets demonstrate BiRank's soundness (fast convergence), efficiency (linear in the number of graph edges), and effectiveness (achieving state-of-the-art in the two real-world tasks).

Index Terms—Bipartite graph ranking, graph regularization, n-partite graphs, popularity prediction, personalized recommendation

1 Introduction

Graphs provide a universal language to represent relationships between entities. In real-world applications, not only should the relationships between entities of the same type be considered, but the relationships between different types of entities should also be modeled. Such

TriRank: explicit aspects join the graph



TriRank: Review-aware Explainable Recommendation by Modeling Aspects·

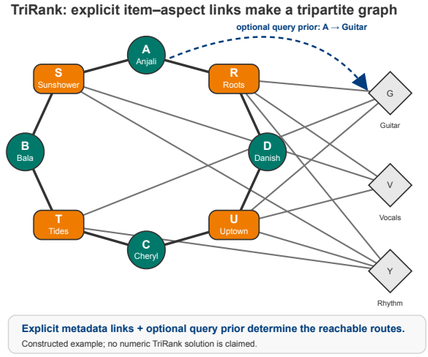
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ABSTRACT

Most existing collaborative filtering techniques have focused on modeling the binary relation of users to items by extracting from user ratings. Aside from users' ratings, their affiliated reviews often provide the rationale for their ratings and identify what aspects of the item they cared most about. We explore the rich evidence source of aspects in user reviews to improve top-N recommendation. By extracting aspects (*i.e.,* the specific properties of items) from textual reviews, we enrich the user–item binary relation to a user–item–aspect ternary relation. We model the ternary relation as a heterogeneous tripartite graph, casting the recommendation task as one of vertex ranking. We devise a generic algorithm for ranking on tripartite graphs — TriRank — and specialize it for personalized recommendation. Experiments on two public review datasets show that it consistently outperforms state-of-the-art methods. Most importantly, TriRank endows the recommender system with a higher degree interest (*e.g.,* products, businesses, movies, *etc.*). Among the various recommendation techniques, collaborative filtering (CF) is widely used, due to its effectiveness in providing personalized recommendation based on the wisdom of the crowds. By and large, existing CF techniques have focused on modeling user–item relations, such as ratings. As a single rating only indicates a user's overall satisfaction for the item, it is hard to infer the actual rationale of the rating. Different users may care about different aspects of the same item. For example, in the restaurant domain, one user may give a 5-star rating for food quality, while another user may give the same rating, but due to favorable ambiance. Since existing CF techniques largely lack such fine-grained analysis, they can fail to accurately model a target user's interests.

Aside from ratings, most Web 2.0 systems also give users the opportunity to leave reviews on the provided items. These reviews often justify a user's rating, offering the underlying reasons for the rating by discussing the specific

Alternate **user, item and aspect scores** through degree-normalized links, anchored by query priors.



Constructed music metadata: Roots/Uptown have Guitar, Vocals, Rhythm; Sunshower has Vocals/Rhythm; Tides has Guitar/Rhythm.

An explicitly supplied **Guitar prior** offers a route to Tides alongside listening evidence.

He, X., Chen, T., Kan, M.-Y., & Chen, X. (2015). *TriRank: Review-aware explainable recommendation by modeling aspects*. In *Proceedings of the 24th ACM International Conference on Information and Knowledge Management* (pp. 1661–1670). ACM.

Break — 5 minutes

WHILE YOU PAUSE · RECOMMENDATION HISTORY

2017 · Pinterest's Pixie

Pixie walked Pinterest's pin–board graph to find related items in real time. Shared boards supplied paths between pins—not just a popularity count (Eksombatchai et al., 2017).

When does a shared graph path reveal relevance—and when might it amplify a community's existing tastes?



▶ Start Timer

Eksombatchai, C., Jindal, P., Liu, J. Z., Liu, Y., Sharma, R., Sugnet, C., Ulrich, M., & Leskovec, J. (2017). *Pixie: A system for recommending 3+ billion items to 200+ million users in real-time*. arXiv. https://doi.org/10.48550/arXiv.1711.07601

03 Learned Propagation

Graph structure becomes a shared embedding operator.



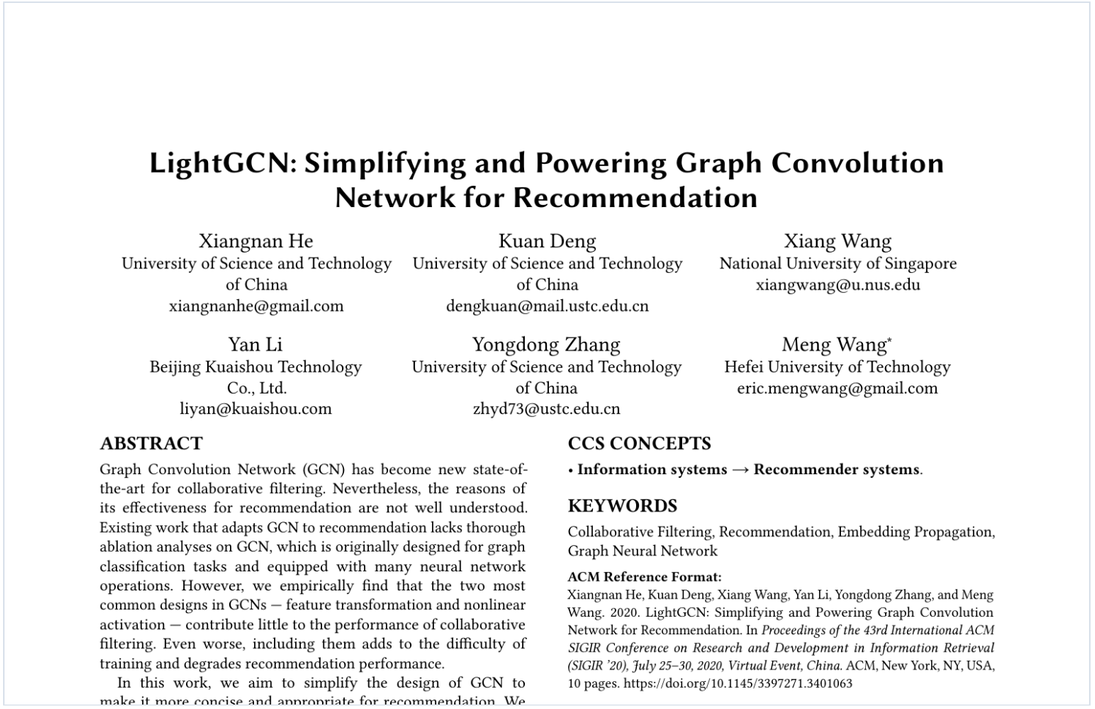
LightGCN: graph-aware collaborative filtering

Matrix factorization: learn user and item vectors, then score their dot product.

NGCF bridge: graph neighbours enrich representations with learned transformations and nonlinearities.

LightGCN: retain normalized neighbour aggregation. Remove feature transforms and nonlinear activations.

Learn the initial embeddings; combine layers and train for ranking.



LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation

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ABSTRACT

Graph Convolution Network (GCN) has become new state-of-the-art for collaborative filtering. Nevertheless, the reasons of its effectiveness for recommendation are not well understood. Existing work that adapts GCN to recommendation lacks thorough ablation analyses on GCN, which is originally designed for graph classification tasks and equipped with many neural network operations. However, we empirically find that the two most common designs in GCNs — feature transformation and nonlinear activation — contribute little to the performance of collaborative filtering. Even worse, including them adds to the difficulty of training and degrades recommendation performance.

In this work, we aim to simplify the design of GCN to make it more concise and appropriate for recommendation. We

CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

Collaborative Filtering, Recommendation, Embedding Propagation, Graph Neural Network

ACM Reference Format:

Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. 2020. LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20), July 25–30, 2020, Virtual Event, China.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3397271.3401063

He, X., Deng, K., Wang, X., Li, Y., Zhang, Y., & Wang, M. (2020). *LightGCN: Simplifying and powering graph convolution network for recommendation*. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 639–648). ACM.

LightGCN: normalized neighbour aggregation

$$e_u^{(k+1)} = \sum_{i \in N_u} \frac{1}{\sqrt{d_u d_i}} e_i^{(k)}.$$

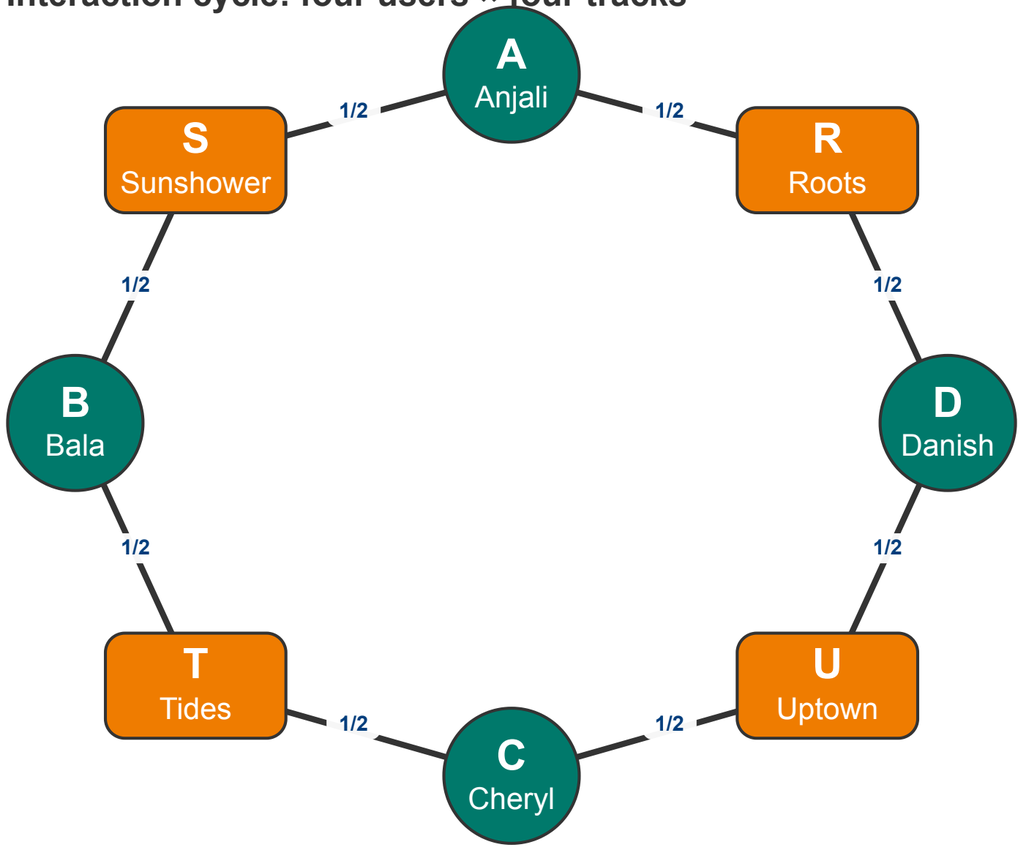
On the original graph:

$$\frac{1}{\sqrt{2 \cdot 2}} = \frac{1}{2}.$$

$$e_A^{(k+1)} = \frac{1}{2} e_R^{(k)} + \frac{1}{2} e_S^{(k)}.$$

Items use the analogous update from user embeddings.
Every node reads the **previous layer**.

Interaction cycle: four users × four tracks



Users = teal circles; tracks = orange rounded rectangles. Every node has degree 2.

One synchronous layer

Illustrative 1D initial embeddings

Node	A	B	C	D
$e_u^{(0)}$	1	3	5	-1

Node	R	S	T	U
$e_i^{(0)}$	2	4	6	-2

$$e_A^{(1)} = \frac{1}{2}\left(\underbrace{2}_{e_R^{(0)}} + \underbrace{4}_{e_S^{(0)}}\right) = 3.$$

$$e_S^{(1)} = \frac{1}{2}\left(\underbrace{1}_{e_A^{(0)}} + \underbrace{3}_{e_B^{(0)}}\right) = 2.$$

Layer 1, all nodes

Users A,B,C,D	Items R,S,T,U
3, 5, 2, 0	0, 2, 4, 2

Next layer uses layer 1:

$$e_A^{(2)} = \frac{1}{2}\left(\underbrace{0}_{e_R^{(1)}} + \underbrace{2}_{e_S^{(1)}}\right) = 1.$$

Layer combination and unseen-item scores

Why combine layers? Different information.

Layer 0: Anjali's learned initial embedding — identity and interaction signal from training.

Layer 1: one-hop neighbourhood information through **Roots and Sunshower**.

For this $K = 1$ example, retain **both**:

$$e_A = \frac{1}{2}\left(\underbrace{1}_{e_A^{(0)}} + \underbrace{3}_{e_A^{(1)}}\right) = 2.$$

Using only layer 1 drops the layer-0 contribution.

With $K = 3$, combine all four layers:

$$e_u = \sum_{k=0}^{3} \alpha_k \, e_u^{(k)}.$$

Layers **0, 1, 2, 3**: original, one-, two-, three-hop signals — not just layers 2 and 3.

Uniform choice: $\alpha_k = 1/(K+1)$; $K = 1$: $1/2$, $K = 3$: $1/4$.

Combine items the same way

$$e_T = \frac{1}{2}(6 + 4) = 5, \qquad e_U = \frac{1}{2}(-2 + 2) = 0.$$

Score unseen items using combined embeddings

$$\hat{y_{AT}} = 2 \times 5 = 10, \qquad \hat{y_{AU}} = 2 \times 0 = 0.$$

Tides ranks above Uptown for these inputs. Filter observed Roots and Sunshower.

The equal-seed walk tied T,U. These illustrative embeddings are **not evidence that LightGCN is better**.

Propagate, combine, score, train

```python
# Pseudocode: sparse normalized adjacency S_hat
layers = [E0]  # trainable user + item embeddings
for k in range(K):
    layers.append(S_hat @ layers[-1])
E = sum(layers) / (K + 1)
users, items = split_users_items(E)
scores = users @ items.T
```

All nodes update together through the sparse matrix. Training gradients flow back to **E0**.

For Anjali: positive **S**, sampled unobserved **U**.

$$\Delta = \hat{y}_{AS} - \hat{y}_{AU}, \qquad L = -\log \sigma(\Delta) + \lambda \| E^{(0)} \|^2.$$

Held-out interactions stay outside the training graph. Filter observed items at recommendation time.



Bandit Time: Graph-learning check

Fix the training graph and layer weights. In this LightGCN model, which parameters does BPR update?

A. Only the final combined embeddings; the initial embeddings stay fixed after initialization.

B. The initial user and item embeddings, through gradients from all propagated layers.

C. A separate embedding table at each layer, updated independently before layer averaging.

D. The initial embeddings and the degree-normalized edge coefficients, jointly optimized by BPR.

Answer in TempoQuiz and explain where the gradient flows.

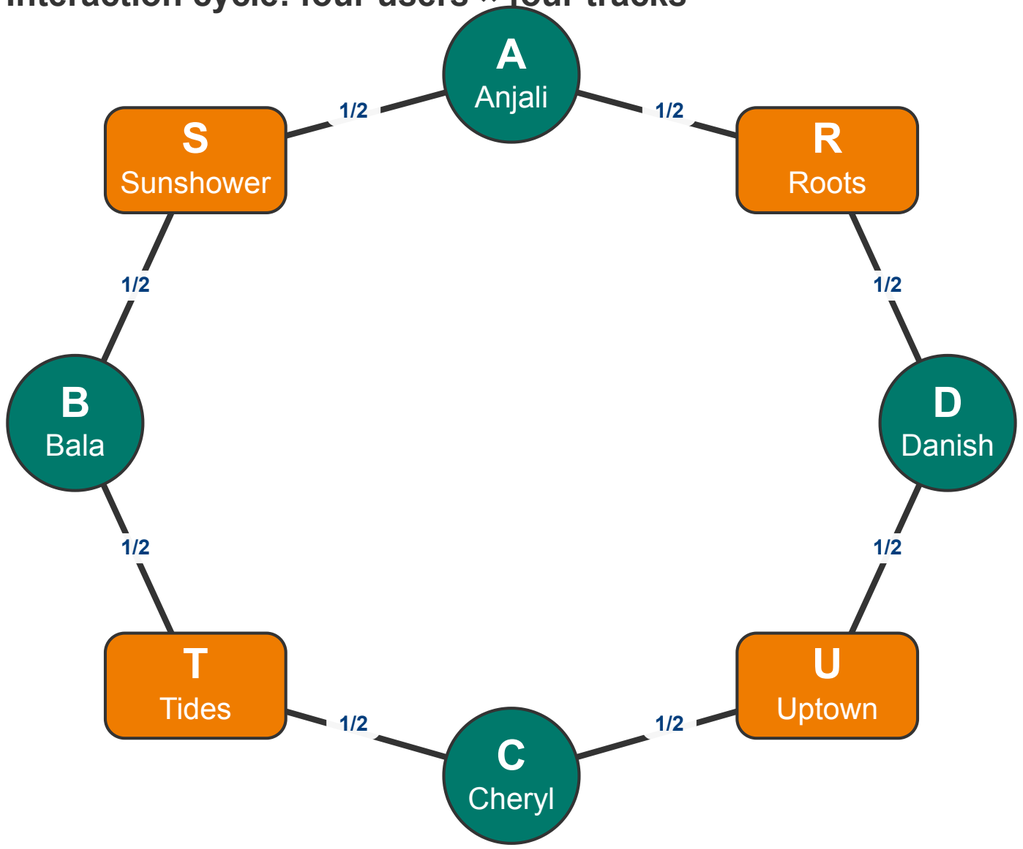
04 Feedback and Personalization

Longer propagation changes whose signal survives.



Mixing can forget the starting user

Interaction cycle: four users × four tracks



Users = teal circles; tracks = orange rounded rectangles. Every node has degree 2.

For this regular graph, the ordinary stationary mass is **1/8 per node**.

Global teleportation makes convergence possible, but the same global operator has the same stationary ranking for every start.

A finite walk retains depth-dependent seed information. A personalised restart keeps a user-specific prior.

Deep embedding propagation can also blur distinctions. This is related to, but not identical with, probability mixing.

👥 One new edge, whose exposure changes?

Add Cheryl–Roots to the music graph

Groups of 2–3: recompute the symmetric normalized weights for C–T, C–U and the new C–R.

Constructed scenario, not a paper result:
NDCG rises 0.30 → 0.34; long-tail exposure falls 25% → 10%.

Output: three weights, one affected route, and one measurable guardrail. Would you deploy?

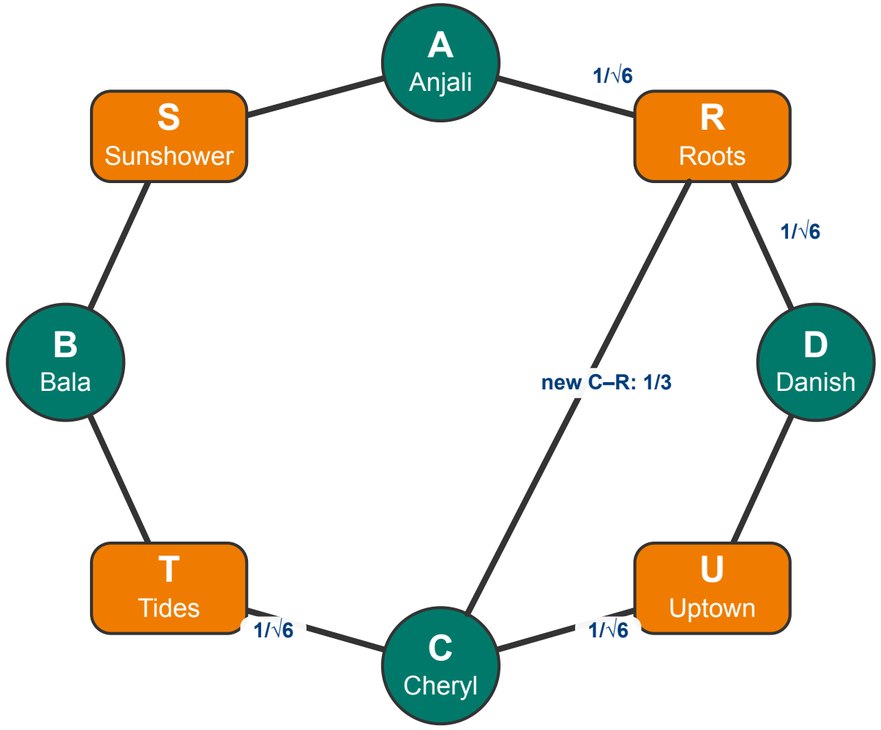
05:00

▶ Start Timer



New degrees change existing messages

Adding Cheryl–Roots changes five normalized edge weights



After addition: deg(C)=deg(R)=3. Audit changed exposure and relevance separately.

After C–R: $d_C = d_R = 3$.

$$w_{CT} = w_{CU} = \frac{1}{\sqrt{3 \cdot 2}} = \frac{1}{\sqrt{6}}.$$

$$w_{CR} = \frac{1}{\sqrt{3 \cdot 3}} = \frac{1}{3}.$$

R–A and R–D also become $1/\sqrt{6}$. Existing messages change even where no new edge appeared.

Example guardrail: long-tail share of eligible top-K impressions, with relevance monitored separately.

Community amplification and feedback

Observed interactions reflect what users had an opportunity to hear.

> ⚠ **Warning**
>
> More exposure can produce more interactions, stronger graph signal and still more exposure. Similar communities may become harder to escape.

Log exposure as well as interactions. Audit coverage and subgroup effects. Preserve a route for unfamiliar but eligible tracks.



Summary

Graph evidence needs an explicit operator and an evaluation goal.



🔑 Connections can reveal and reinforce taste

Representation: what do nodes and edges mean?

Propagation: probability, coupled score or embedding?

Personalization: seed, prior or learned parameters?

Evaluation: relevance gain with which exposure guardrail?

Next week: **multi-objective recommendation**.



NEXT WEEK PRE-LECTURE EXERCISE

Challenge an improvement through one lens

Canvas · Post + peer critique · 15 minutes
Complete by Mon, 19 Oct 2026 · 23:59 SGT

Your post · 100–140 words: name a surface, propose a graph-recommender change, and state a primary metric and benefit. Choose one **primary lens** below.

♠ Stable requirements

Preserve eligibility, important item attributes, consistency, or an understandable reason for inclusion.

♦ Discovery

Help users find relevant unfamiliar items; examine variety and long-tail exposure.

♥ Social influence

Use social evidence helpfully while limiting conformity pressure or unwanted disclosure.

Peer reply · 50–70 words: quote a classmate's claim. Through their lens, give a case where the primary metric improves but the intended outcome worsens. Name a measurable guardrail to detect this.

Canvas: initial post by **Sun, 18 Oct · 23:59 SGT**; peer reply by the Monday deadline. These are design lenses, not labels for people.

Next lecture: turn "better" into explicit objectives and defensible trade-offs.





NEXT WEEK PREVIEW — WEEK 10

Multi-Objective Recommendation

Image generated with GPT